

EDITH — Even Dead I'm The Hero

A Multi-Agent AI Assistant Framework for Autonomous Software Development and Intelligent Task Automation

Overview

EDITH (Even Dead I'm The Hero) is an advanced multi-agent AI assistant designed to move beyond traditional chatbot architectures. Instead of relying on a single AI model, EDITH integrates multiple AI providers, intelligent routing, autonomous software generation, and modular plugins into a unified desktop environment.

Objectives

Build a modular multi-agent AI framework; integrate multiple AI providers; automatically recover from provider failures; generate complete software projects autonomously; support intelligent routing and task automation; provide a local-first, extensible architecture.

System Architecture

React HUD Interface → Flask + Socket.IO → Core Intelligence Layer (AI Router, AI Engine, Project Agent, Memory Manager, Self-Healing Engine) → Plugin Layer (Voice, Vision, Weather, Email, Maps, Utilities, System Monitor) → AI Providers (Gemini, Groq, OpenRouter, Mistral, Ollama).

Key Features

Multi-Agent Architecture, Automatic Provider Failover, Intelligent Routing, Autonomous Software Generation, Document Analysis, Image & Video Generation, Voice Interaction, Vision Processing, Email Integration, Maps & Weather, Live System Monitoring, Virus Shield, Plugin-Based Architecture, Real-Time Streaming, Conversation Memory.

Technologies Used

Python, Flask, Flask-SocketIO, React, HTML, CSS, JavaScript, Google Gemini, Groq, OpenRouter, Mistral, Ollama, Requests, psutil, BeautifulSoup.

Engineering Challenges

Coordinating multiple AI providers, managing asynchronous workflows, handling provider outages gracefully, maintaining project state, building reusable modular components, and streaming long-running operations efficiently.

Results

EDITH demonstrates autonomous software generation, multi-agent orchestration, intelligent routing, provider failover, real-time communication, voice interaction, vision analysis, plugin integration, and live system monitoring.

Future Roadmap

Model Context Protocol (MCP), Docker deployment, Kubernetes support, multi-user authentication, persistent memory, mobile application, automated testing, additional AI providers, and advanced reasoning agents.

Developer

Kamalesh JK AI & Data Science Student GitHub: <https://github.com/kamaleshjk102007-dot/edith>
Demo: <https://youtu.be/oIT5tEaFCSU>

Conclusion

EDITH demonstrates how multiple specialized AI agents, intelligent routing, resilient provider management, and modular software engineering principles can create a powerful next-generation

AI assistant platform.